

# Relaxation gap of a trajectory-averaged fermionic reset circuit

Class-A fermionic adaptive-circuit project

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We relate the relaxation of the outcome-averaged density matrix to the single-particle projector product  $A$  for one measurement cycle. The dimensionless channel gap is  $g = 1 - r_{\text{MB}}$ , distinct from the logarithmic decay rate  $\Delta = -\log r_{\text{MB}}$ . For perfect occupation resets in a fixed repeated schedule, the charge-neutral and parity-even many-body gaps equal  $g_C = 1 - \rho(A)^2$ . Higher-order correlations cannot relax more slowly than the two-point function. The result allows noncommuting, support-truncated Wannier measurements.

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## I. MEASUREMENT CYCLE AND TWO-POINT RELAXATION

After discarding outcomes and fresh ancillary modes, the physical-layer channel fills lower-band overcomplete Wannier (OW) modes and empties upper-band modes, as in Bhuiyan, Pan, and Jian (BPJ), Appendix D [1]. We retain exact discrete resets in their prescribed order.

Let  $\{\psi_i, \psi_j^\dagger\} = \delta_{ij}$  for  $N$  fermion modes. At step  $j$ , measure  $\hat{\chi}_j = \sum_i v_{ji} \psi_i$ , with  $v_j^\dagger v_j = 1$  and  $\hat{N}_j = \hat{\chi}_j^\dagger \hat{\chi}_j$ . Averaging the measurement and perfect feedback gives the physical-layer channels

$$\begin{aligned}\tilde{\mathcal{E}}_{j,0}(\hat{\rho}) &= \hat{\chi}_j \hat{\rho} \hat{\chi}_j^\dagger + (\hat{1} - \hat{N}_j) \hat{\rho} (\hat{1} - \hat{N}_j), \\ \tilde{\mathcal{E}}_{j,1}(\hat{\rho}) &= \hat{\chi}_j^\dagger \hat{\rho} \hat{\chi}_j + \hat{N}_j \hat{\rho} \hat{N}_j.\end{aligned}\quad (1)$$

The target occupation is  $s_j = 0$  or  $1$ . The projector terms include measurement dephasing; no separate dephasing step is added. One cycle is the composition  $\tilde{\mathcal{E}} = \tilde{\mathcal{E}}_{M,s_M} \circ \dots \circ \tilde{\mathcal{E}}_{1,s_1}$ , with step 1 first. Distinct OW modes need not be orthogonal, so this order matters.

Write  $\bar{G}_{ab} = \text{Tr}(\bar{\rho} \hat{\psi}_a^\dagger \hat{\psi}_b)$  and set  $P_j = v_j v_j^\dagger$ ,  $Q_j = \hat{1} - P_j$ . A reset removes correlations with the measured mode and sets its occupation:

$$\bar{G}' = Q_j \bar{G} Q_j + s_j P_j. \quad (2)$$

The part of the initial two-point function surviving one cycle is therefore propagated by

$$A = Q_M \dots Q_1, \quad \bar{G}_{n+1} = A \bar{G}_n A^\dagger + B. \quad (3)$$

The source  $B$  depends on the reset targets, not on the input state. For deviations from a stationary correlation matrix,  $\delta G_{n+1} = A \delta G_n A^\dagger$ . Column vectorization gives

$A^* \otimes A$ , whose eigenvalues are  $a_i a_j^*$  if  $a_i$  are those of  $A$ . Define the spectral radius  $r = \rho(A) = \max_i |a_i|$ . No eigenvalue of  $A$  is excluded:  $A$  is not the many-body channel, and stationarity is the affine condition  $G_{\text{ss}} = A G_{\text{ss}} A^\dagger + B$ , not an eigenvalue 1 of  $A$ . The two-sided action accounts for the square: the largest covariance multiplier is  $r^2$ . For  $0 < r < 1$ ,

$$g_C = 1 - r^2, \quad \Delta_C = -2 \log r. \quad (4)$$

Here  $g_C$  is dimensionless and  $\Delta_C$  is per complete cycle. The spectral radius, not the largest singular value, sets asymptotic decay. In the computation we diagonalize both invariant hard-wall blocks of the ordered product and take the largest eigenvalue modulus across both. A unit-modulus eigenvalue would give  $g_C = 0$ ; none is discarded.

## II. FROM TWO-POINT DECAY TO THE MANY-BODY GAP

Fix an invariant operator sector containing the stationary state. For a mixing channel,  $\tilde{\mathcal{E}}(\hat{\rho}_{\text{ss}}) = \hat{\rho}_{\text{ss}}$  provides a simple eigenvalue 1. A nonstationary eigenoperator satisfies  $\tilde{\mathcal{E}}(\hat{X}_\nu) = \mu_\nu \hat{X}_\nu$ , so its amplitude after  $n$  cycles is  $\mu_\nu^n$ . Define

$$\begin{aligned}r_{\text{MB}} &= \max_{\nu \text{ nonstationary}} |\mu_\nu|, & g_{\text{MB}} &= 1 - r_{\text{MB}}, \\ \Delta_{\text{MB}} &= -\log r_{\text{MB}}, & g_{\text{MB}} &= 1 - e^{-\Delta_{\text{MB}}}.\end{aligned}\quad (5)$$

The absolute channel gap is the deficit from unity in *modulus*, not  $\min_\nu |1 - \mu_\nu|$  for complex eigenvalues. Only the stationary mode is removed; another unit-modulus mode would obstruct mixing. The gap and decay rate agree only to leading order near zero.

Covariance closure alone gives a bound. The Heisenberg channel preserves the span of  $\hat{1}$  and all  $\hat{\psi}_a^\dagger \hat{\psi}_b$ , so its restriction has eigenvalues 1 and  $a_i a_j^*$ . These also belong to the full channel spectrum. In any mixing sector containing these operators,

$$r_{\text{MB}} \geq r^2, \quad g_{\text{MB}} \leq g_C, \quad \Delta_{\text{MB}} \leq \Delta_C. \quad (6)$$

This is the spectral-inclusion bound: a positive two-point rate alone cannot exclude a slower higher-order correlation.

For  $A^\dagger u = a^* u$ , the centered occupation

$$\hat{O}_u = \sum_{ab} u_a^* u_b (\hat{\psi}_a^\dagger \hat{\psi}_b - (G_{ss})_{ab} \hat{1}) \quad (7)$$

obeys  $\tilde{\mathcal{E}}^\dagger(\hat{O}_u) = |a|^2 \hat{O}_u$ . Choosing  $|a| = r$  explicitly realizes the slow neutral density mode.

Choose a basis starting with  $\hat{\chi}_j$  and let  $\hat{B}$  contain only orthogonal modes. For even total operators, Eq. (1) gives

$$\begin{aligned} \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{B}) &= \hat{B} & (\hat{B} \text{ even}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{\chi}_j^{(\dagger)} \hat{B}) &= 0 & (\hat{B} \text{ odd}), \\ \tilde{\mathcal{E}}_{j,s}^\dagger(\hat{N}_j \hat{B}) &= s \hat{B} & (\hat{B} \text{ even}). \end{aligned} \quad (8)$$

A reset erases a measured leg; an occupation pair contributes  $s$  and reduces the degree by two. Thus  $2p$ -point functions couple only to themselves and lower orders, without assuming Wick factorization.

Charge-neutral correlations take the form

$$F_{I;J}^{(p)} = \text{Tr} \left( \hat{\rho} \hat{\psi}_{i_1}^\dagger \cdots \hat{\psi}_{i_p}^\dagger \hat{\psi}_{j_p} \cdots \hat{\psi}_{j_1} \right), \quad (9)$$

where  $I, J$  are increasing  $p$ -element sets. The order- $2p$  part propagates creation indices with  $A$  and annihilation indices with  $A^*$ . Antisymmetry gives the one-cycle block

$$H_p[I, J; K, L] = \det A[I, K] \det A[J, L]^*. \quad (10)$$

Cauchy–Binet composes the minors of the  $Q_j$  in the actual order. The hierarchy is block triangular in  $p$ ; lower-order terms do not change its eigenvalues. The multipliers are

$$\Lambda_{I,J} = \prod_{i \in I} a_i \prod_{j \in J} a_j^*, \quad |I| = |J| = p. \quad (11)$$

The independent sets may overlap, allowing the same label on both legs. These  $\sum_p \binom{N}{p}^2 = \binom{2N}{N}$  moments span all neutral operators. Schur triangularization gives the same products for nondiagonalizable  $A$ .

The constant ( $p = 0$ ) supplies eigenvalue 1. Every  $p \geq 1$  product obeys  $|\Lambda_{I,J}| \leq r^{2p} \leq r^2$ , with equality at  $p = 1$  for  $I = J = \{i_*\}$ ,  $|a_{i_*}| = r$ . No higher neutral correlation is slower; the stationary eigenvalue is simple and

$$\begin{aligned} g_{q=0} &= g_C = 1 - \rho(A)^2, \\ \Delta_{q=0} &= \Delta_C = -2 \log \rho(A). \end{aligned} \quad (12)$$

### III. SECTORS AND EARLIER RESULTS

Charge neutral means  $[\hat{N}_F, \hat{\rho}] = 0$ , where  $\hat{N}_F = \sum_i \hat{\psi}_i^\dagger \hat{\psi}_i$ ; it permits mixtures of number sectors. Definite-charge reset Kraus operators preserve this condition while changing particle number. Maximal mixing and fixed-number Slater preparations both lie in this sector.

With  $p$  creations and  $\ell$  annihilations, the even-sector argument gives products of  $p + \ell$  factors. Every nonconstant even product has at least two, while the neutral sector attains  $r^2$ . Hence  $g_{\text{even}} = g_{q=0}$  and  $\Delta_{\text{even}} = \Delta_{q=0}$ . To include odd operators, use the similarity  $\mathcal{S}(\hat{O}) = (-1)^{\hat{N}_F} \hat{O}$ . It changes the sign of the odd-Kraus sandwich in  $\mathcal{S} \tilde{\mathcal{E}}_j^\dagger \mathcal{S}$ ; on odd operators that sign cancels the fermionic reordering sign and restores Eq. (8). Odd products then have maximal modulus  $r$ , attained with one factor. For the specified Kraus extension on unrestricted operator space,

$$g_{\text{full}} = 1 - r, \quad \Delta_{\text{full}} = -\log r = \Delta_C/2. \quad (13)$$

Physical density matrices are parity even. Thus the plotted  $g_C$  is the relevant many-body channel gap for our states, not merely a bound. Earlier plots of  $\Delta_C$  show the corresponding decay rate instead; neither quantity is the occupation-spectrum gap around  $1/2$ .

The continuous-time counterpart is Barthel–Zhang, Sec. III.6, Proposition 7 and Eq. (63): the covariance generator is a block of the many-body Liouvillian [2]. Their Sec. III.5, Proposition 6 and discussion after Eq. (57d), distinguish unrestricted and two-point rates. Equation (6) is the discrete spectral-inclusion bound (also Ref. [3], Eq. (4)); equality (12) requires our additional exact-reset argument.

BPJ Appendix D, Eqs. (D34)–(D40), instead treats effective continuous-time bulk dissipation rates [1], not this ordered-reset identity. Nonnormal transients and Jordan prefactors remain possible: our result fixes an asymptotic rate, not a size-uniform mixing or trajectory-purification time. It does not prove a thermodynamic extrapolation. Extra dephasing, imperfect feedback, or randomized schedules requires a new higher-order analysis.

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- [1] A. Bhuiyan, H. Pan, and C.-M. Jian, *Phys. Rev. Research* **8**, 023147 (2026), Appendix D.  
 [2] T. Barthel and Y. Zhang, *J. Stat. Mech.* (2022) 113101, Secs. III.5–III.6; [arXiv:2112.08344v5](#), pp. 14–16.

- [3] A. Bhuiyan, *A spectral bound on relaxation of the trajectory-averaged circuit*, project note (September 29, 2026), Eq. (4); [local PDF](#).